YOUSEF ALSAYED ABDELFATTAH

Junior AI Engineer
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SUMMARY

AI Engineer with hands-on experience building LLM-powered applications, multi-agent systems, RAG pipelines, and ML solutions. Experienced in model evaluation, fine-tuning, backend development, and deploying AI workflows using Python, PyTorch, LangChain, and Docker.

WORK EXPERIENCE

AI Engineering Intern — *Remote*
Arche Robotics — *Jul 2026 – Present*
- Currently developing a monitor agent that parses and analyzes live robot log streams to detect anomalies and surface them to the engineering team in real time

Programming Instructor — *Cairo, Egypt*
iSchool – English-speaking coding education platform for K-12 and adult learners — *May 2026 – Present*
- Taught programming fundamentals to 30+ students across 2 beginner cohorts, covering core concepts such as logic, control flow, variables, functions, and basic problem solving

Military Service — *Egypt*
Egyptian Armed Forces Engineering Authority – Compulsory national service — *Jan 2025 – Mar 2026*

PROJECTS

End-to-End Multi-Agent SDR Pipeline
OpenAI Agents SDK · Groq · Pydantic · Langfuse · SendGrid
- Built an end-to-end sales automation pipeline (3 drafter agents, 1 manager, 1 emailer) with Pydantic validation and Langfuse/OpenTelemetry tracing, reducing the manual workflow to 4 automated stages across 6+ integrated tools

Advanced Engineering Team — Multi-Agent Code Generator
CrewAI · Groq · Docker · LangFuse
- Built a 4-agent CrewAI pipeline with a Docker-based sandboxed execution environment, producing a Python module, Gradio interface, and unit tests from natural-language specs with full LangFuse tracing

RAG-AI-Agent — Domain-Specialized Document Intelligence Agent
LangChain · ChromaDB · Hugging Face · Groq · Streamlit
- Built an agentic LangChain RAG system with 4 integrated capabilities: semantic document retrieval, web search, voice playback, and document collection management in a unified Streamlit interface

EgyptianSavoryScout — Fine-Tuned LLM Restaurant Recommender
HuggingFace Transformers · PyTorch · Qwen 1.5B · scikit-learn · FastAPI
- Fine-tuned Qwen 1.5B on a 1,000+ entry dataset and built a TF-IDF + rating ranker for recommendations across 7 Egyptian cities

EDUCATION

Misr University for Science and Technology — *Cairo, Egypt*
Bachelor of Science, Mechatronics, Robotics & Automation Engineering — *June 2024*
Graduation Project: R2-WAITER — autonomous restaurant waiter robot built with 4 core technologies: ROS 1, Nav2, Gazebo, and PyQt5
- Designed the navigation stack with Nav2 (SLAM mapping, AMCL localization, Kalman/EKF sensor fusion across LIDAR and depth-camera data) and a layered software architecture simulated in Gazebo, plus a secure per-order locking system with a live PyQt5 UI

CERTIFICATIONS

Machine Learning Specialization — DeepLearning.AI — *Aug 2026*
AI Engineer Agentic Track: The Complete Agent and MCP Course — Udemy — *Jun 2026*
AI Engineer Core Track — Udemy — *May 2026*
AWS AI Practitioner Challenge — Udacity — *Apr 2026*
CS50's Introduction to Artificial Intelligence with Python — Harvard University — *Nov 2024*
CS50's Introduction to Computer Science — edX — *Dec 2023*

SKILLS

Programming: Python, C++, C, PLC Programming
ML/Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, QLoRA, PEFT
LLM/GenAI: RAG, embeddings, semantic search, LLM evaluation, prompt engineering
Agentic AI: LangGraph, LangChain, OpenAI Agents SDK, AutoGen, MCP, tool calling
Backend, Deployment & Observability: FastAPI, Docker, async Python, Git/GitHub, Langfuse, LangSmith, OpenTelemetry, Streamlit, Gradio
Languages: English (C2), Arabic (Native)